MIDDLESEX SCHOOL | MATHEMATICS

Math 32 Course at a Glance

Trigonometry Fundamentals

Topical units are not numbered because instructional order may vary. Objectives are grouped by curriculum priority; supporting skills are intentionally omitted from this summary.

Trigonometric Foundations

REQUIRED	
M32-FND-001	I can use the Pythagorean theorem to determine an unknown side of a right triangle.
M32-FND-002	I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.
M32-FND-003	I can convert angle measures between degrees and radians.
M32-FND-004	I can determine coterminal and reference angles for a given angle.
M32-FND-005	I can calculate arc length from a central angle and radius.
M32-FND-006	I can determine exact coordinates on the unit circle.
M32-FND-007	I can evaluate all six trigonometric functions exactly for common angles.
M32-FND-008	I can solve right triangles using trigonometric ratios.
M32-FND-009	I can apply right-triangle trigonometry to contextual problems.
M32-FND-010	I can evaluate trigonometric functions from a point on a circle or terminal side.
M32-FND-011	I can solve a basic trigonometric equation for all angles in a specified interval.

Periodic Functions and Modeling

REQUIRED	
M32-GRF-001	I can graph basic sine and cosine functions using key features.
M32-GRF-002	I can identify amplitude, midline, period, and phase shift from an equation or graph.
M32-GRF-003	I can graph transformed sine and cosine functions.
M32-GRF-004	I can write sine and cosine equations that represent a periodic graph.
M32-GRF-005	I can construct a sinusoidal model from contextual information or data.
M32-GRF-006	I can interpret the parameters and outputs of a sinusoidal model in context.
M32-GRF-007	I can graph tangent, secant, and cosecant functions with their asymptotes.

Inverse Functions and Trigonometric Equations

REQUIRED	
M32-INV-001	I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.
M32-INV-002	I can explain how restricted domains make inverse trigonometric functions possible.
M32-INV-003	I can solve trigonometric equations exactly or approximately on a specified interval.
M32-INV-004	I can express all solutions of a trigonometric equation in general form.
M32-INV-005	I can evaluate compositions of trigonometric and inverse trigonometric functions.

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Trigonometry Fundamentals

Non-Right Triangles

REQUIRED	
M32-TRI-002	I can solve a non-right triangle using the Law of Sines.
M32-TRI-003	I can determine the number of possible triangles in the ambiguous SSA case.
M32-TRI-004	I can solve a non-right triangle using the Law of Cosines.
M32-TRI-005	I can select and apply an appropriate method to solve a triangle in context.

Trigonometric Identities

REQUIRED	
M32-IDN-001	I can rewrite trigonometric expressions using reciprocal and quotient identities, including `tan(theta) = sin(theta)/cos(theta)`.
M32-IDN-002	I can rewrite trigonometric expressions using the Pythagorean identities.
M32-IDN-003	I can use even/odd and cofunction identities to rewrite trigonometric expressions.
M32-IDN-004	I can verify a trigonometric identity using the required identity families.

Additional Trigonometric Identities

EXTENSION	
M32-IDX-001	I can apply double-angle identities.
M32-IDX-002	I can apply half-angle identities.
M32-IDX-003	I can apply angle sum and difference identities.
M32-IDX-004	I can determine exact trigonometric values using extension identities.

Triangle Area from SAS

EXTENSION	
M32-TRX-001	I can calculate the area of a triangle from two sides and the included angle.

Polar Coordinates

REQUIRED	
M32-POL-001	I can plot and identify points in polar coordinates.
M32-POL-002	I can represent a polar point using equivalent coordinate pairs.
M32-POL-003	I can convert points between polar and rectangular coordinates.

Polar Functions

EXTENSION	
M32-PLX-001	I can graph and interpret basic polar functions.
M32-PLX-002	I can convert equations between polar and rectangular forms.

Vectors

EXTENSION	
M32-VEC-001	I can represent a vector geometrically and in component form.
M32-VEC-002	I can add, subtract, and multiply vectors by scalars.
M32-VEC-003	I can determine vector magnitude, direction, and unit vector.
M32-VEC-004	I can resolve a vector into horizontal and vertical components.
M32-VEC-005	I can perform basic operations with vectors in three dimensions.

Parametric Equations

EXTENSION	
M32-PAR-001	I can graph a plane curve defined by parametric equations with its orientation.
M32-PAR-002	I can eliminate a parameter to obtain a rectangular equation.
M32-PAR-003	I can create or interpret parametric equations for motion in the plane.